Roll No							

End Semester Examination 2024

Name of the Course: DIPLOMA(CSE)

Semester: 4TH

Name of the Paper: COMPUTER NETWORKS

Paper Code: DTCS 403

Time: 3 Hour's

Maximum Marks: 100

Note:

(i) All Questions are compulsory.
(ii) Answer any two sub questions among a,b and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(10 X2 = 20 Marks)**	**CO 1**
(a)	Distinguish between different network classifications based on their geography, such as LAN (Local Area Network), MAN (Metropolitan Area Network), and WAN (Wide Area Network). What are the main characteristics and applications of each type of network?	
(b)	Explain the role of centralized management in networked environments. How do networks enable organizations to manage software, maintain the network, and back up data centrally?	
(c)	Define network and explain its significance in modern computing. How do human networks and computer networks differ, and how are they interconnected?	
Q2	**(10 X2 = 20 Marks)**	**CO 2**
(a)	Discuss the suitability of each topology for different network environments and applications, considering factors such as scalability, fault tolerance, and ease of maintenance?	
(b)	Explain the concept of IP addressing and its importance in network communication. How are IP addresses assigned and classified into different classes?	
(c)	Discuss the role of each network control device in managing and directing network traffic, improving network performance, and enabling connectivity between different network segments and protocols?	
Q3	**(10 X2 = 20 Marks)**	**CO 3**
(a)	Compare and contrast different types of guided media, including twisted pair, coaxial cable, and optical fiber. What are the main characteristics, advantages, and limitations of each type?	
(b)	Explain the concept of wireless communication and the use of radio frequencies for transmitting data wirelessly. What are the main communication bands used in wireless communication, and how are they allocated for different purposes?	

(c)	Describe the bands used in cellular telephony and the organization of cellular networks into cells. How do mobile phones establish and maintain calls using cellular networks, including the process of handover between cells?	
Q4	**(10 X2 = 20 Marks)**	**CO 4**
(a)	Explain the concept of the OSI (Open Systems Interconnection) Reference Model and its importance in standardizing network communication protocols. What are the seven layers of the OSI model, and what functions does each layer perform?	
(b)	Describe the functions of the Transport layer in the OSI model, including segmentation, error detection, and flow control. What are the main protocols and standards associated with the Transport layer, such as TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)?	
(c)	Compare and contrast the OSI and TCP/IP reference models in terms of their structure, layering, and protocol suite. What are the advantages and limitations of each model in describing network communication?	
Q5	**(10 X2 = 20 Marks)**	**CO 5**
(a)	Compare and contrast SLIP (Serial Line Internet Protocol) and PPP (Point-to-Point Protocol) in terms of their features, advantages, and limitations. What are the main differences between SLIP and PPP, and why is PPP considered more versatile and reliable?	
(b)	Explain the concept of IP addressing and the structure of IP addresses. How are IP addresses assigned to devices on a network, and what are the different IP address classes (A, B, C, D, E)?	
(c)	Compare and contrast TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) in terms of their features, reliability, and use cases. When would you choose TCP over UDP, and vice versa?	